

Lab 1, Act 2: Derive

Turning a candidate shape into a testable equation

NRES 746 – name: _____ group: _____

Picking up where we left off

In Act 1, your group diagnosed the tadpole predation data as **hump-shaped** and named the Ricker function, $y = a x e^{-bx}$, as a plausible candidate. Today we make that candidate concrete: we'll pick specific numbers for a and b , evaluate the function, and use calculus to find exactly where it peaks – then check whether that matches what you eyeballed last time.

Before we get there, a quick warm-up on the algebra tool we'll lean on all semester: logarithms and exponents.

Part 1: Log & exponent warm-up (individual, ~10 min)

Useful rules:

$$\log(xy) = \log x + \log y \qquad \log\left(\frac{x}{y}\right) = \log x - \log y \qquad \log(x^k) = k \log x$$

$$e^{\ln x} = x \qquad \ln(e^x) = x \qquad \ln x \approx 2.303 \times \log_{10} x$$

1a. Apply the rules. Simplify each:

$$\log_{10}(4) + \log_{10}(25) = \underline{\hspace{2cm}}$$

$$\log_{10}(1000) - \log_{10}(10) = \underline{\hspace{2cm}}$$

$$\log_{10}(2^{10}) = \underline{\hspace{2cm}} \quad (\text{you'll need } \log_{10}(2) \approx 0.301)$$

1b. Solve for x:

$$2^x = 32 \Rightarrow x = \underline{\hspace{2cm}} \qquad 3^x = 81 \Rightarrow x = \underline{\hspace{2cm}} \qquad 10^x = 0.001 \Rightarrow x = \underline{\hspace{2cm}}$$

$$e^x = 20 \Rightarrow x = \underline{\hspace{2cm}}$$

1c. Linearizing an exponential decay. This is the trick that will let us fit exponential-type models with plain linear regression later this semester. Take natural logs of both sides of $y = ae^{-bx}$ and simplify using the rules above. Show your work:

$$\underline{\hspace{10cm}}$$
$$\underline{\hspace{10cm}}$$

$$\ln y = \underline{\hspace{4cm}}$$

What form is this, as a function of x ? What do the slope and intercept correspond to, in terms of a and b ?

$$\underline{\hspace{10cm}}$$

Now check your answer against a small numeric example: $y = 10 e^{-0.5x}$ (so $a = 10$, $b = 0.5$). Fill in the table below.

Useful values: $e^{-1} \approx 0.368$, $e^{-2} \approx 0.135$, $e^{-3} \approx 0.050$, $\ln(10) \approx 2.303$

x	$y = 10e^{-0.5x}$	$\ln(y)$
0		
2		
4		
6		

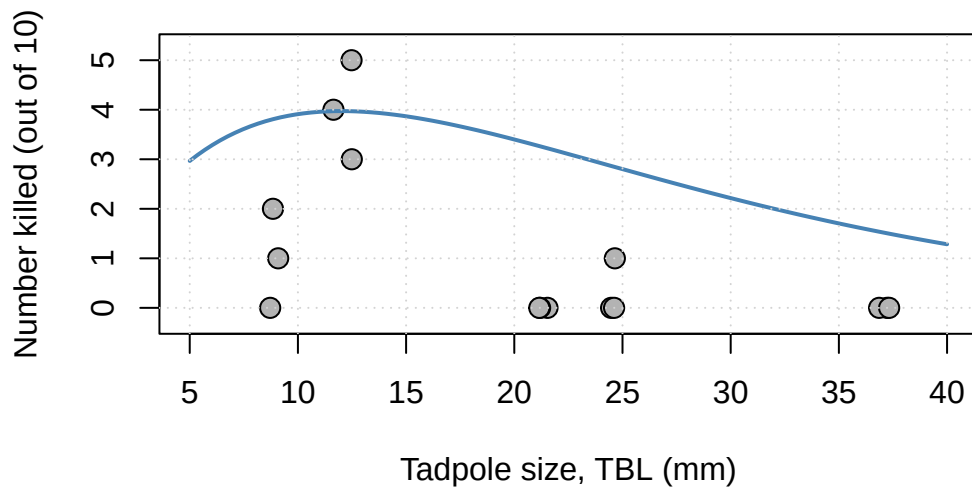
Compute the change in $\ln(y)$ each time x increases by 2: _____. Does it match what you predicted for the slope above? _____

Part 2: Derive properties of the Ricker function (individual, ~20 min)

Step 1: Today's candidate. Let's put specific numbers on the Ricker shape from Act 1:

$$y = axe^{-bx}, \quad a = 0.9, \quad b = \frac{1}{12} \text{ mm}^{-1}$$

(These are illustrative, chosen to roughly match the pattern you saw – not a formal fit. We'll learn how to actually *fit* a curve to data soon!)



Step 2: Evaluate by hand. Using $a = 0.9$ and $b = 1/12$, compute y at three tadpole sizes. *Useful values:* $e^{-0.5} \approx 0.607$, $e^{-1} \approx 0.368$, $e^{-2} \approx 0.135$.

$$y(6) = \quad y(12) = \quad y(24) =$$

Do these three values rise then fall, consistent with the hump shape? _____

Step 3: Take the derivative. Find dy/dx for $y = axe^{-bx}$. This is a *product* of two pieces of x , so you'll need the product rule together with the chain rule for the exponential term.

$$\text{Reminders: } \frac{d}{dx}(uv) = u'v + uv' \quad \frac{d}{dx}e^{g(x)} = g'(x)e^{g(x)}$$

Show your work:

$$\frac{dy}{dx} = \frac{\quad}{\quad}$$

Step 4: Find the critical point. Set $dy/dx = 0$ and solve for the critical point x^* , using our candidate's $b = 1/12$. (Hint: think about which factor(s) in your derivative could actually equal zero.)

Show your work:

$$x^* = \quad \text{mm}$$

Step 5: Interpret. In a sentence, what does x^* mean ecologically – what is happening to a tadpole of exactly this size?

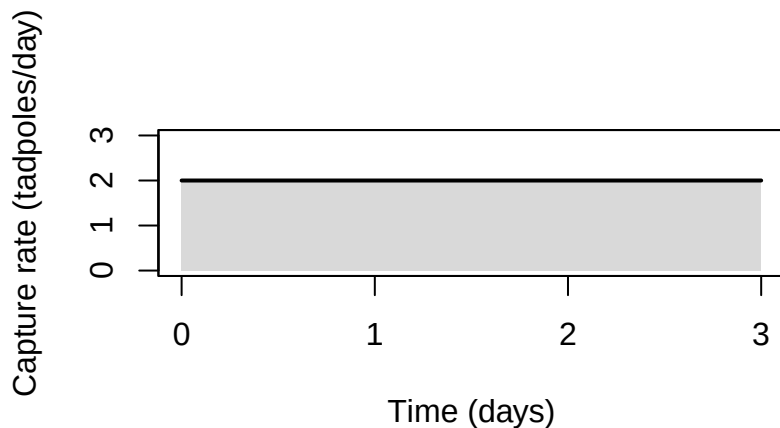
Look back at the curve you sketched by hand in Act 1. Does the calculus-derived peak, $x^* = \quad$ mm, land close to where you drew your peak? If not, what might explain the difference?

Part 3: A first taste of integration (individual, ~10 min)

Calculus has two main tools: derivatives (rates of change, which we just used to find a peak) and **integrals** (accumulation – adding up a rate over some interval to get a total). We'll use this idea heavily once we get to probability and likelihood.

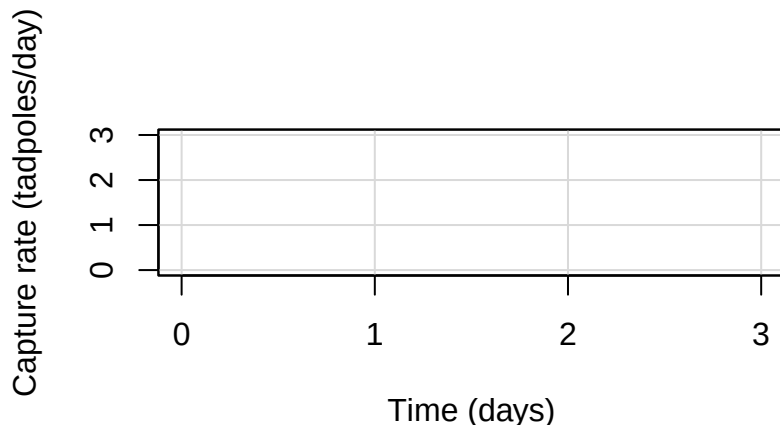
Worked example. Suppose – just for this simple example, forget the Ricker curve – that dragonfly larvae capture tadpoles at a *constant* rate of $r = 2$ tadpoles/day for the entire $T = 3$ -day trial. The total number killed is rate $\times$ time:

$$\text{Total killed} = r \times T = 2 \times 3 = 6 \text{ tadpoles}$$



Geometrically, that's just the **area of the shaded rectangle**: base (time) $\times$ height (rate). "Area under the rate curve = cumulative total" is exactly the idea behind integration – we'll just need it for curves that aren't flat lines.

Your turn. Now suppose the capture rate isn't constant – it ramps up *linearly*, from 0 tadpoles/day at $t = 0$ to 2 tadpoles/day at $t = 3$ (dragonflies take some time to find the tadpoles). Sketch this rate on the blank axes below, shade the region representing the total killed, and compute the total using the area of a triangle.



Total killed = $\frac{1}{2} \times \text{base} \times \text{height} = \underline{\hspace{2cm}}$

(This example is deliberately simple and separate from the Ricker curve – it's just to plant the "area = total" idea. We'll connect integration to our tadpole model later in the semester.)

Group discussion (15 min)

Compare your work with your group:

- Walk through your derivative, term by term. Did everyone get the same expression for dy/dx ? Resolve any disagreements about the product rule or chain rule steps.
- Did everyone arrive at the same critical point, x^* ? If not, find where the calculations diverge.
- Does the calculus-derived peak you each calculated match where you sketched the peak by hand in Act 1? As a group, discuss: is a mismatch a problem with the calculus, a problem with our illustrative a, b choice, or just normal eyeballing imprecision?
- Compare your Part 1 and Part 3 answers too, if time allows – these are foundational tools we'll reuse constantly.

Revise and reflect (individual, ~15 min)

Revise any of your answers above based on the group discussion. Then answer:

1. Write your final expression for dy/dx and the critical point x^* in your own words. What does x^* represent ecologically?

2. Now that you have a specific predicted peak size, how confident are you that this is a real ecological effect and not just noise in a small dataset? What would you want to know to decide?

3. Muddiest point: what's the one thing from this part of the lab you'd like to see explained again?
